

FlowTSFM: Turning Encoder Depth into Quantile Transport

Bahaeddine Abdesslem^{1,3} Shifeng Xie^{1,2} Zehao Xiao¹
 Youssef Attia El Hili^{1,4} Ambroise Odonnat^{1,5} Jianfeng Zhang⁶ Lujia Pan⁶
 Keli Zhang¹ Malik Tiomoko¹

¹Huawei Noah’s Ark Lab, Paris, France

²LIPADE, Université Paris Cité, Paris, France

³École Polytechnique, France

⁴Centre de Recherche en Informatique, Mines Paris, PSL University, France

⁵IRISA, Université Rennes 2, Inria, France

⁶Huawei Noah’s Ark Lab, Shenzhen, China

Abstract

Encoder-based time series foundation models (TSFMs) typically rely on deep stacks of independently parameterized Transformer layers, where only the final forecast is supervised and intermediate representations have no explicit predictive role. We introduce FlowTSFM, an encoder architecture that interprets depth as a recurrent transport process: a single Transformer block is iteratively applied with shared parameters, while a quantile-flow objective supervises intermediate states along a prescribed trajectory from a prior distribution toward the final forecast. The objective combines pinball forecasting loss with path-level position matching. With only 38.8M parameters, FlowTSFM achieves competitive performance on GIFT-Eval and TIME, remaining within 1.8–4.6% MASE of stronger baselines while using approximately 3× fewer parameters than a 12-layer Chronos-2 model (119.5M). Beyond accuracy, we introduce CosMean, a scale-free diagnostic measuring whether recurrent updates consistently align toward the final prediction. Under a matched intermediate-state probing protocol, FlowTSFM achieves a CosMean score of 0.919 compared with 0.350 for Chronos-2, suggesting that recurrent parameter sharing combined with path supervision is associated with substantially more structured predictive trajectories at a favorable accuracy-efficiency trade-off.

1 Introduction

Time series forecasting is an important task across several domains, such as finance [21], healthcare [10], cloud operations [12], and industrial monitoring [24]. Forecasts influence consequential decisions (support planning, resource allocation, risk management) yet real-world signals differ widely in scale, sampling rate, noise, and temporal structure [19], motivating general-purpose probabilistic forecasters, often built on patch-based representations [18], that transfer across datasets and domains rather than requiring a separate model per series [3].

Recent time-series foundation models achieve strong forecasting performance through encoder-based, decoder-based, and recurrent architectures [3, 6, 19, 23]. However, in encoder-based models, depth remains largely an internal computation: intermediate states have no explicit forecasting role, while supervision ultimately targets the resulting forecast. Generative flow matching offers a contrasting view in which intermediate computation describes transport along a path between distributions [14], but this idea has primarily been developed for sample-space generative modeling. It therefore remains unclear whether encoder depth in probabilistic forecasting can instead be organized as a structured transport process over predictive quantiles.

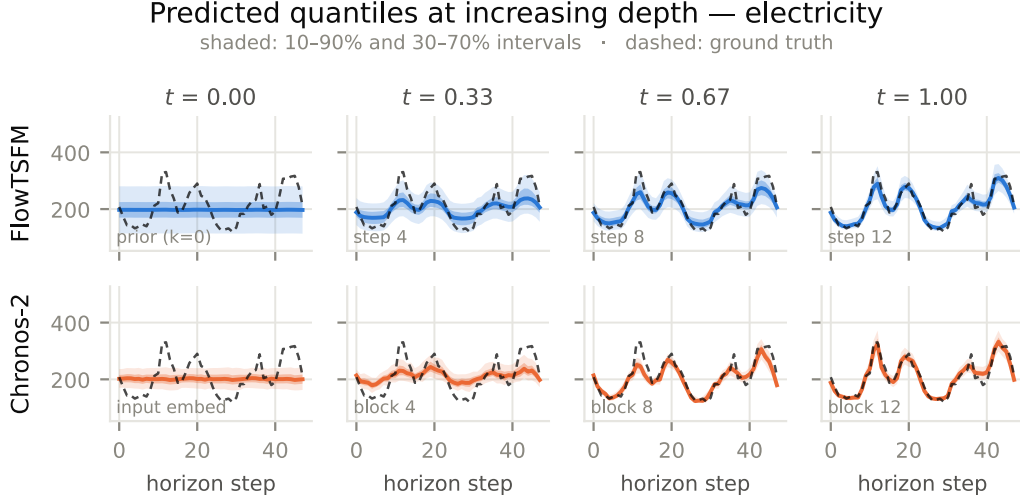


Figure 1: **Depth-wise quantile evolution on a held-out Electricity-H series.** Columns correspond to increasing normalized depth $t = \frac{k}{K}$. In each panel, the solid line is the median forecast, the darker and lighter bands are the 30–70% and 10–90% predictive intervals, and the dashed line is the ground truth. FlowTSMF (top row) already forms a coherent forecast at early exits and refines it smoothly across depth. Chronos-2 (bottom row), probed with the same frozen output head, shows weaker early forecast structure and larger late-stage corrections.

We introduce FlowTSMF, a forecasting framework that interprets encoder depth as quantile transport. FlowTSMF replaces the multiple independent encoder layers with a single shared block applied recurrently through an ODE-like residual update [5], in the spirit of recent work on weight-tied recursive computation [11]. We further introduce a quantile-flow objective that supervises intermediate predictions along a prescribed trajectory toward the terminal forecast. Unlike generative flow matching, our formulation does not learn a sample-space velocity field; the recurrence evolves in latent space while its decoded predictive quantiles are directly constrained along depth.

Empirically, FlowTSMF (38.8 parameters) comes within 1.8–4.6% MASE of the strongest baselines on GIFT-Eval [1] and TIME [20], while using roughly $3\times$ fewer parameters than a 12-layer Chronos-2 (119.5M). Under a matched trajectory protocol, FlowTSMF reaches a mean cosine alignment (CosMean) of 0.919 toward its terminal forecast, compared to 0.350 for an equivalently probed Chronos-2 (Figure 1). Together, these results indicate that intermediate encoder computation can be given an explicit geometric role at a favorable accuracy–efficiency trade-off, rather than as a strict improvement over larger, non-recurrent encoders. Concretely, our contributions are:

- (i) a recurrent encoder architecture that replaces independent Transformer layers with a single weight-tied block, cutting parameter count by roughly $3\times$ relative to a comparable 12-layer model;
- (ii) a quantile-flow training objective that supervises intermediate decoder exits along a prescribed source-to-forecast trajectory, giving encoder depth an explicit predictive role;
- (iii) We employ CosMean[16], a scale-free diagnostic for whether recurrent updates advance toward the terminal forecast, and a matched trajectory protocol for applying it across architecturally different models.

2 Methodology

Notation. We write $[n] = \{1, \dots, n\}$. The normalized input is x and the observation mask is m ; the encoder E maps the pair (x, m) to the initial latent state $h^0 = E(x, m) \in \mathbb{R}^{T \times d_{\text{model}}}$, where d_{model} is the latent space dimension. Let $\mathcal{M} \subseteq [T]$ be the set of forecast positions scored by the loss, and y_s the target at position $s \in \mathcal{M}$.

We discretize depth into K steps: for $k \in \{0, \dots, K\}$ we set $t_k = \frac{k}{K}$, with step size $\Delta t = \frac{1}{K}$. Each $h^k \in \mathbb{R}^{T \times d_{\text{model}}}$ is the latent state at depth k , and the decoder D_ϕ maps it to the decoded field $q^k = D_\phi(h^k) = (q_{s,j}^k) \in \mathbb{R}^{T \times Q}$, whose entry $q_{s,j}^k$ is the prediction at position s for quantile level τ_j , $j \in [Q]$. We call each selected depth k an *exit*.

Finally, $r = (r_j)_{j \in [Q]}$ is the analytic *source* attached to the quantile levels, defined by $r_j = \text{arcsinh}(\Phi^{-1}(\tau_j))$ and broadcast across all forecast positions. This analytic source represents the decoding of the first hidden layer that we obtain directly after the embedding of the time series. The analytic source represents the default quantile predictions of the model, which in this case we will use the quantiles of a normal distribution. Therefore we use $r_j = \text{arcsinh}(\Phi^{-1}(\tau_j))$ rather than $r_j = \Phi^{-1}(\tau_j)$ because we further use arcsinh [4] normalization.

Recurrent encoder architecture A conventional K -layer encoder computes $h^{k+1} = B_{\theta_k}(h^k)$ with independently learned θ_k . FlowTSFM instead reuses one shared velocity field F_θ at every depth:

$$h^{k+1} = h^k + \Delta t F_\theta(h^k, e_t(t_k), e_{\text{depth}}(K)), \quad k = 0, \dots, K-1. \quad (1)$$

Here $e_t(t_k)$ encodes the normalized depth and $e_{\text{depth}}(K)$ encodes the total step budget, allowing the same recurrent block to adapt to different unroll lengths. Both F_θ and the output chart D_ϕ are shared across depth. This update resembles an ODE integration in the latent space[5]

Quantile-flow objective Let ℓ_{pin}^k be the mean *pinball loss* at exit k , i.e., the standard proper loss for quantile forecasting, and let $\pi^k = (1 - t_k)r + t_k \text{sg}(q^K)$ be the path target interpolating between the source and the terminal prediction where sg denotes the stop-gradient operator, which leaves its input unchanged in the forward pass but blocks gradient propagation during backpropagation. During training, we supervise a fixed number of intermediate exits: at each step, a fixed-size subset $\mathcal{E}' \subseteq \{1, \dots, K-1\}$ is sampled uniformly. For these sampled exits, we minimize

$$\mathcal{L} = \ell_{\text{pin}}^K + \sum_{k \in \mathcal{E}'} \left[\lambda_{\text{ds}} t_k \ell_{\text{pin}}^k + \lambda_{\text{qf}} \|q^k - \pi^k\|_{\mathcal{M}}^2 \right] + \lambda_0 \|q^0 - r\|_{\mathcal{M}}^2. \quad (2)$$

Here $\|\cdot\|_{\mathcal{M}}^2$ is the mean squared deviation over scored positions and quantile levels. The first term ℓ_{pin}^K optimizes the terminal forecast q^K ; the depth-scaled intermediate pinball term keeps early exits predictive; the path-matching term makes intermediate decoded quantiles follow a prescribed source-to-endpoint trajectory; and the last term anchors the learned initial exit q^0 to the source r . The stop-gradient ensures that this auxiliary path supervision shapes the trajectory without directly moving the terminal target. Exact definitions are deferred to Appendix B.

3 Results

Experimental setting. The accuracy test makes use of an 8192-point context, a patch size of 16, $K = 10$, 99 quantiles, and a 50/50 combination of domain-balanced GIFT-Eval-Pretrain data and kernel-generated synthetic data. It is trained for 100,000 AdamW steps and is assessed via the Seasonal-Naive-normalized geometric aggregations as specified by GIFT-Eval and TIME. For trajectory evaluation, we use the released 12-layer Chronos-2.

Metrics. Following directional diagnostics for rectified and self-consistent flows [9, 16], we use only mean cosine alignment between each decoded update and the remaining displacement to q^K .

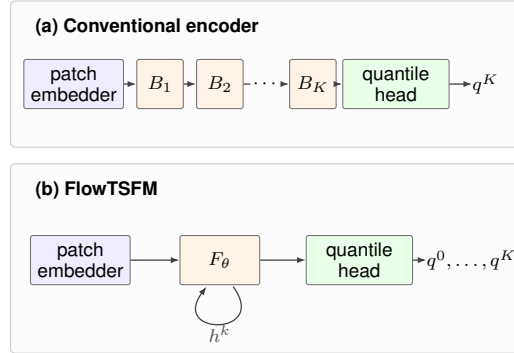


Figure 2: Conventional depth (top) uses independent blocks $B_1, \dots, B_K$. FlowTSFM (bottom) replaces them with one recurrent block applied repeatedly, producing intermediate forecasts $q^0, \dots, q^K$. See Appendix B for implementation details.

Table 1: Seasonal-Naive-normalized forecasting accuracy on the two standard benchmarks. GIFT-Eval contains 97 configurations and TIME contains 98 tasks. Lower is better.

Model	GIFT MASE ↓	GIFT CRPS ↓	TIME MASE ↓	TIME CRPS ↓	Parameters
FlowTSFM (base)	0.729	0.500	0.694	0.580	38.8M
Sundial (base)	0.750	0.559	0.758	0.663	128.3M
Tirex	0.716	0.488	0.683	0.573	35.3M
Moirai-2	0.728	0.516	0.703	0.588	11.4M
Timesfm 2.0	0.758	0.550	0.718	0.620	498.8M
Chronos-2	0.697	0.485	0.662	0.556	119.5M

Table 2: Mean cosine alignment between local trajectory velocities and the endpoint chord on GIFT-Eval. Dataset-macro averages weight each of the 55 datasets equally, whereas path-weighted averages weight every valid test trajectory equally. Higher is better.

Model	Dataset macro ↑	Path weighted ↑
FlowTSFM	0.919	0.933
Chronos-2	0.350	0.317

Intuitively, this metric is high when the forecast trajectory advances toward its terminal prediction rather than wandering or reversing:

$$\text{CosMean}(q) = \frac{1}{K-1} \sum_{k=0}^{K-2} \frac{\langle q^{k+1} - q^k, q^K - q^k \rangle}{\|q^{k+1} - q^k\|_2 \|q^K - q^k\|_2}. \quad (3)$$

Inner products flatten $(s, j) \in \mathcal{M} \times [Q]$. The final update is omitted because its cosine is one by construction. Hence $\text{CosMean}(q) \in [-1, 1]$, with larger values indicating more endpoint-directed transport.

4 Discussion and conclusion

FlowTSFM makes two linked changes to the encoder-only architecture: it replaces the K separate layers with K instances of an ODE-like field in latent space, and it substitutes terminal-only training with a quantile-flow objective which gives the recurrent interior a well-defined purpose by constraining it. The reduction in the number of parameters is a direct result of weight tying. The geometric findings indicate that the objective has successfully assigned a coherent role to the repeated computation and produced more directionally aligned quantile transport.

The evidence remains preliminary: accuracy and geometry use different single-run checkpoints; Chronos-2’s early states were not trained or supervised in the same way as in our current setting. Furthermore, ablation studies for the new objective, along with tests with different values of K , are still pending. The narrower result is that decoded path supervision yields a much more structured recurrent trajectory at comparable terminal quality.

References

- [1] T. Aksu, G. Woo, J. Liu, et al. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.
- [2] A. Alexandrov, K. Benidis, M. Bohlke-Schneider, V. Flunkert, J. Gasthaus, T. Januschowski, D. C. Maddix, S. Rangapuram, D. Salinas, J. Schulz, L. Stella, A. C. Türkmen, and Y. Wang. GluonTS: Probabilistic time series models in Python. *Journal of Machine Learning Research*, 21(116):1–6, 2020.
- [3] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang. Chronos: Learning the language of time series. In *International Conference on Machine Learning (ICML)*, 2024.
- [4] A. F. Ansari, O. Shchur, J. Küken, et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.
- [5] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [6] A. Das, W. Kong, R. Sen, and Y. Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024.
- [7] K. Frans, D. Hafner, S. Levine, and P. Abbeel. One step diffusion via shortcut models. *arXiv preprint arXiv:2410.12557*, 2024.
- [8] J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès. One-step generative modeling via Wasserstein gradient flows. *arXiv preprint arXiv:2605.11755*, 2026.
- [9] X. Han, J. Hu, and L.-P. Liu. Self-consistent flow: Unifying velocity and endpoint prediction for rectified flow models. *arXiv preprint arXiv:2607.12171*, 2026.
- [10] A. E. W. Johnson, L. Bulgarelli, L. Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023.
- [11] A. Jolicoeur-Martineau. Less is more: Recursive reasoning with tiny networks. *arXiv preprint arXiv:2510.04871*, 2025.
- [12] A. Joosen, A. Hassan, M. Asenov, R. Singh, L. Darlow, J. Wang, and A. Barker. How does it function? characterizing long-term trends in production serverless workloads. In *Proceedings of the ACM Symposium on Cloud Computing*, pages 443–458, 2023.
- [13] T. Kim, J. Kim, Y. Tae, C. Park, J.-H. Choi, and J. Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations (ICLR)*, 2022.
- [14] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- [15] C. Liu, T. Aksu, J. Liu, X. Liu, H. Yan, Q. Pham, S. Savarese, D. Sahoo, C. Xiong, and J. Li. Moirai 2.0: When less is more for time series forecasting. *arXiv preprint arXiv:2511.11698*, 2025.
- [16] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023.
- [17] Y. Liu, G. Qin, Z. Shi, Z. Chen, C. Yang, X. Huang, J. Wang, and M. Long. Sundial: A family of highly capable time series foundation models. In *International Conference on Machine Learning (ICML)*, 2025.
- [18] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with Transformers. In *International Conference on Learning Representations (ICLR)*, 2023.
- [19] P. Podest, M. Pichler, E. Bürger, L. Zólyomi, B. Voggenberger, W. Berghammer, D. Klotz, S. Böck, G. Klambauer, and S. Hochreiter. TiRex-2: Generalizing TiRex to multivariate data and streaming. *arXiv preprint arXiv:2607.01204*, 2026.
- [20] Z. Qiao, S. Pan, A. Wang, V. Zhukova, Y. Liu, X. Jiang, Q. Wen, M. Long, M. Jin, and C. Liu. It’s TIME: Towards the next generation of time series forecasting benchmarks. In *International Conference on Machine Learning (ICML)*, 2026.

- [21] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu. Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. *Applied Soft Computing*, 90:106181, 2020.
- [22] Y. Wen, W. M. Gifford, C. Reddy, L. M. Nguyen, J. Kalagnanam, and A. A. Julius. Revisiting the generic transformer: Deconstructing a strong baseline for time series foundation models. *arXiv preprint arXiv:2602.06909*, 2026.
- [23] G. Woo, C. Liu, A. Kumar, C. Xiong, S. Savarese, and D. Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024.
- [24] P. Yan, A. Abdulkadir, P. Luley, M. Rosenthal, G. A. Schatte, B. F. Grewe, and T. Stadelmann. A comprehensive survey of deep transfer learning for anomaly detection in industrial time series: Methods, applications, and directions. *IEEE Access*, 12:3768–3789, 2024.

A Related work

Generative flow matching [14] learns a time-dependent vector field that transports a simple source distribution toward a data distribution along a prescribed probability path. Training samples a time and a state on this path and regresses the model toward the corresponding conditional velocity, avoiding likelihood evaluation and full ODE simulation during training [5, 14]. Recent work studies shortcut parameterizations and the compression of such transport paths into very few, or even one, generation steps [7, 8]. In parallel, time-series foundation models pretrain probabilistic forecasters on heterogeneous corpora: patch-based architectures underpin Chronos, TimesFM, Moirai, and Chronos-2 [3, 4, 6, 23], while TiRex-2 demonstrates the continued relevance of recurrent designs [19].

B Detailed architecture and training configuration

Figure 3 expands the architecture in Section 2, separating the one-time input construction, the shared recurrent field, and the decoder with its training objectives.

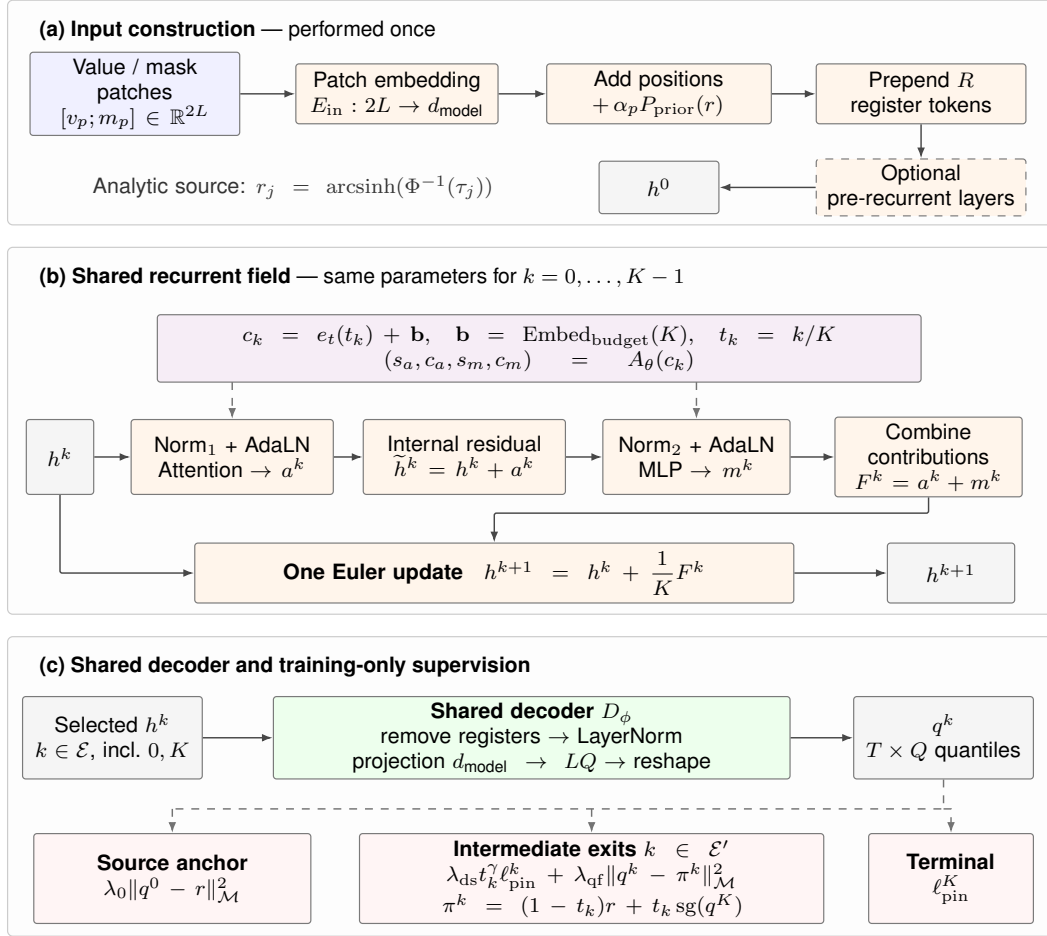


Figure 3: Detailed FlowTSMF computation. (a) Input construction runs once; optional pre-recurrent layers are outside the loop. (b) Attention and MLP contributions form one shared field; only the Euler block updates the recurrent state. (c) The shared decoder also reads h^0 , so q^0 is learned and anchored to r . Dashed arrows denote conditioning or training-only supervision.

Notation and preprocessing. Let x_{raw} be the raw series and m its binary observation mask. Following the instance-normalization principle of ReVIN [13], we compute μ and σ from visible context values only and set $x = \text{arcsinh}((x_{\text{raw}} - \mu)/\sigma)$; masked values do not contribute to these statistics, and targets y_s use the same coordinates. Writing $[n] = \{1, \dots, n\}$, the shared decoder produces $q^k = D_\phi(h^k) = (q_{s,j}^k)_{s \in [T], j \in [Q]} \in \mathbb{R}^{T \times Q}$ at every selected exit $k \in \mathcal{E} \subseteq \{0, \dots, K\}$, while losses retain only $s \in \mathcal{M} \subseteq [T]$. The source $r = (r_j)_{j \in [Q]}$, with $r_j = \text{arcsinh}(\Phi^{-1}(\tau_j))$, is broadcast across positions. In particular, $q^0 = D_\phi(h^0)$ is learned and is anchored to r by the final term of Eq. 2; it is not defined as r . We write $\mathcal{E}' \subseteq \{1, \dots, K-1\}$ for the intermediate supervised exits sampled during training; at each training step, a fixed-size subset is drawn uniformly from the available interior exits. Superscripts index recurrent depth, not exponentiation.

Loss definitions. For any exit k , the compact terms in Eq. 2 are

$$\begin{aligned} \ell_{\text{pin}}^k &= \frac{1}{|\mathcal{M}|Q} \sum_{s \in \mathcal{M}} \sum_{j=1}^Q \rho_{\tau_j}(y_s - q_{s,j}^k), & \rho_{\tau}(u) &= \max\{\tau u, (\tau - 1)u\}, \\ \|A\|_{\mathcal{M}}^2 &= \frac{1}{|\mathcal{M}|Q} \sum_{s \in \mathcal{M}} \sum_{j=1}^Q A_{s,j}^2. \end{aligned}$$

Thus $\|q^k - \pi^k\|_{\mathcal{M}}^2$ is deterministic position matching to the linear source–endpoint path. Since $t_0 = 0$, $\pi^0 = r$, so the last term of Eq. 2 is exactly the learned exit-0 source anchor. Stop-gradient blocks gradients through the terminal value used to construct π^k ; the recurrent field and decoder remain shared trainable parameters.

Input and recurrent block. Length- L value patches are concatenated with their union-mask patches and mapped from $2L$ to width D by the input patch embedding. Learned positional embeddings are added in the reported configurations. A learned projection $P_{\text{prior}}(r)$ of the source vector is added to each patch in proportion to its prediction-mask fraction α_p , after which R learned register tokens are prepended. This injection conditions the latent computation on the transport origin; it does not replace the learned exit-0 readout $q^0 = D_\phi(h^0)$. Optional pre-recurrent Transformer layers, when configured, act once at this point rather than inside the recurrent loop. For $t_k = k/K$ and $\Delta t = 1/K$, FlowTSFM adds a depth-time embedding and a step-budget embedding, then produces four adaptive-normalization controls:

$$c_k = e_t(t_k) + e_{\text{depth}}(K), \quad (s_a, c_a, s_m, c_m) = A_\theta(c_k). \quad (4)$$

Writing $\text{mod}(z; s, c) = z \odot (1 + c) + s$, the attention and MLP sublayers compose a single velocity before the recurrent state is updated:

$$a^k = \text{MHA}_\theta(\text{mod}(\text{N}_1(h^k); s_a, c_a)), \quad (5)$$

$$\tilde{h}^k = h^k + a^k, \quad (6)$$

$$m^k = \text{MLP}_\theta(\text{mod}(\text{N}_2(\tilde{h}^k); s_m, c_m)), \quad (7)$$

$$F^k = a^k + m^k, \quad h^{k+1} = h^k + \Delta t F^k. \quad (8)$$

MHA is vanilla non-causal multi-head self-attention with $q/k/v/o$ projections and the patch padding mask. The MLP is $d_{\text{model}} \rightarrow 4d_{\text{model}} \rightarrow d_{\text{model}}$ with GeLU. The intermediate sum $\tilde{h}^k = h^k + a^k$ is internal to the Transformer parameterization of the field; it is not an Euler transition. The only recurrent transition is the final line of Eq. 8. There are no attention, MLP, or residual gates. All block parameters are reused for every k .

Exits and decoder. At every selected exit $k \in \mathcal{E}$, including $k = 0$, register positions are removed, final LayerNorm is applied, and the shared output chart D_ϕ maps each patch from d_{model} to LQ values before reshaping to a $T \times Q$ quantile field. Consequently, $q^0 = D_\phi(h^0)$ is a learned decoder output; because $\pi_{s,j}^0 = r_j$, the λ_0 term anchors this initial field to the analytic source. In the base model, the input embedding and output chart are residual projection blocks. The Chronos-2-I/O variant replaces those two modules with the exact pretrained-compatible Chronos-2 residual blocks; the FlowTSFM recurrent core and any optional pre-recurrent layers remain separate trainable model parameters.

Table 3 reports the settings of the base accuracy run. The base checkpoint supplies the GIFT-Eval and TIME scores. The Chronos-2-I/O checkpoint supplies the controlled trajectory study because it shares Chronos-2’s input/output geometry and the nine common quantile levels used by the matched protocol.

Table 3: Reported FlowTSFM configuration for the base accuracy run.

Setting	Base accuracy run
Context length / patch size	8,192 / 16
d_{model} / attention heads	1,024 / 16
Quantiles / recurrent calls	99 / 12
Register tokens	4
Parameters	38,845,536
Real / kernel-synthetic data	50% / 50%
Nominal prediction-mask ratio	40%
Optimizer	AdamW
precision	bfloat16
Peak / final learning rate	3×10^{-4} / 10^{-5}
Warm-up / weight decay	5k / 0.1
$\lambda_{\text{ds}}, \lambda_{\text{qf}}$	0.5, 0.3

The prior projection is learned jointly with the recurrent block and shared decoder. Its role is to condition the latent starting state; the observable path begins at the learned decoding q^0 , which Eq. 2 anchors to r before supervising the positive-depth intermediate decodings.

Benchmark-score provenance. Table 1 reports MASE and CRPS using each benchmark’s taskwise Seasonal-Naive normalization followed by geometric aggregation. The FlowTSFM entries are loaded from the completed base evaluations. The Chronos-2 GIFT-Eval entries aggregate its officially submitted results over the 97 configurations shared with the Seasonal-Naive baseline; its TIME entries are the benchmark’s reported aggregate over 98 tasks [1, 4, 20].